

Polynomial — Pre-Algebra Ready Type Examples

One Easy, Medium, and Hard sample per Ready Pre-Algebra type, formatted like printable worksheets (shared instructions, graphs, and diagrams).

Adding and subtracting

pa_polynomials_adding_and_subtracting · Pre-Algebra – Beginning Polynomials ·
gen=polynomial_add_subtract

Name: _____ Date: _____

Simplify.

EASY

1. $(2x^2 - x + 2) - (x^2 - x - 3)$

MEDIUM

2. $(4x^3 - 9x^2 + 10x + 3) + (2x^3 - 9x^2 - 10x + 2)$

HARD

3. $(3x^3 + x^2 - 2x - 12) - (21x^3 + 19x^2 + 16x - 22)$

Answer Key

1. **Easy:** $x^2 + 5$

2. **Medium:** $6x^3 - 18x^2 + 5$

3. **Hard:** $-18x^3 - 18x^2 - 18x + 10$

Multiplying

pa_polynomials_multiplying · Pre-Algebra – Beginning Polynomials · gen=polynomial_multiply

Name: _____ Date: _____

Multiply.

EASY

1. $(x)(x + 2)$

MEDIUM

2. $(2x + 3)(5x^3 + 9x + 7)$

HARD

3. $(17x^5 - 8x^3 + 15x^2 - 5x - 14)(10x^4 - 9x + 7)$

Answer Key

1. **Easy:** $x^2 + 2x$

2. **Medium:** $10x^4 + 15x^3 + 18x^2 + 41x + 21$

3. **Hard:** $170x^9 - 80x^7 - 3x^6 + 69x^5 - 68x^4 - 191x^3 + 150x^2 + 91x - 98$

Multi-step equations

pa_equations_multi_step_equations · Pre-Algebra – Equations · gen=multi_step_equations

Name: _____ Date: _____

Solve for x .

EASY

1. $2(x + 3) + 3 = 13$

MEDIUM

2. $4(x + 2) - 12 = 8$

HARD

3. $2(x + 5) + 28 = \frac{212}{5}$

Answer Key

1. **Easy:** 2

2. **Medium:** 3

3. **Hard:** $\frac{11}{5}$

One-step equations, word problems

pa_equations_one_step_word_problems · Pre-Algebra – Equations · gen=wp_one_step_equation

Name: _____ Date: _____

Solve the problem.

EASY

1. Hassan had some money, spent \$15, and has \$55 left. How much did Hassan start with?

MEDIUM

2. Will had some money, spent \$14, and has \$56 left. How much did Will start with?

HARD

3. Tessa shared \$232 equally among 4 friends. How much did each friend receive?

Answer Key

1. **Easy:** \$70
2. **Medium:** \$70
3. **Hard:** \$58

Two-step equations word problems

pa_equations_two_step_word_problems · Pre-Algebra – Equations · gen=wp_two_step_equation

Name: _____ Date: _____

Solve the problem.

EASY

1. 9 less than 5 times a number is 56.
Find the number.

MEDIUM

2. 6 less than 5 times a number is 84.
Find the number.

HARD

3. 15 less than 4 times a number is 21. Find the number.

Answer Key

1. **Easy:** 13
2. **Medium:** 18
3. **Hard:** 9

Divisibility

pa_divisibility · Pre-Algebra – Factors and Exponents · gen=g6_divisibility

Name: _____ Date: _____

Determine if the number is divisible.

EASY

1. Is 15 divisible by 5?

MEDIUM

2. Is 9 divisible by 6?

HARD

3. Is 25 divisible by 5?

Answer Key

1. **Easy:** Yes

2. **Medium:** No

3. **Hard:** Yes

Squares and square roots

pa_squares_and_square_roots · Pre-Algebra – Factors and Exponents · gen=pa_squares_and_square_roots

Name: _____ Date: _____

Evaluate.

EASY

1. What is 3 squared?

MEDIUM

2. What is 25 squared?

HARD

3. $\sqrt{750}$

Answer Key

1. **Easy:** 9

2. **Medium:** 625

3. **Hard:** $5\sqrt{30}$

Multi-step inequalities

pa_multi_step_inequalities · Pre-Algebra – Inequalities · gen=multi_step_inequalities

Name: _____ Date: _____

Solve for x .

EASY

1. $3(x + 1) - 3 \geq -6$



MEDIUM

2. $3(x + 5) - 3 \geq -24$



HARD

3. $4(x + 4) + 12 < \frac{40}{3}$

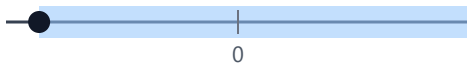


Answer Key

1. **Easy:** $x \geq -2$



2. **Medium:** $x \geq -12$



3. **Hard:** $x < -\frac{11}{3}$



Adding and subtracting

pa_integers_adding_and_subtracting · Pre-Algebra – Integers, Decimals, and Fractions ·
gen=pa_integers_adding_and_subtracting

Name: _____ Date: _____

Evaluate.

EASY

1. $7 - 5$

MEDIUM

2. $\frac{3}{2} - \frac{7}{6}$

HARD

3. $45.56 - (-8.533)$

Answer Key

1. **Easy:** 2

2. **Medium:** $\frac{1}{3}$

3. **Hard:** 54.093

Converting fractions and decimals

pa_converting_fractions_and_decimals · Pre-Algebra – Integers, Decimals, and Fractions ·
gen=converting_fractions_and_decimals

Name: _____ Date: _____

Convert.

EASY

1. Write 0.9 as a fraction.

MEDIUM

2. Write 0.25 as a fraction.

HARD

3. Write $\frac{1}{5}$ as a decimal.

Answer Key

1. **Easy:** $\frac{9}{10}$

2. **Medium:** $\frac{1}{4}$

3. **Hard:** 0.2

Dividing

pa_integers_dividing · Pre-Algebra – Integers, Decimals, and Fractions · gen=g6_integer_divide

Name: _____ Date: _____

Divide.

EASY

1. $-4 \div 7$

MEDIUM

2. $7 \div (-4)$

HARD

3. $-31 \div (-9)$

Answer Key

1. **Easy:** -1

2. **Medium:** -2

3. **Hard:** 3

Factoring

pa_factoring · Pre-Algebra – Integers, Decimals, and Fractions · gen=g6_factoring

Name: _____ Date: _____

Factor.

EASY

1. Write the prime factorization of 49

MEDIUM

2. Write the prime factorization of 21

HARD

3. Write the prime factorization of 102

Answer Key

1. **Easy:** 7^2

2. **Medium:** $3 \cdot 7$

3. **Hard:** $2 \cdot 3 \cdot 17$

Greatest common factor

pa_greatest_common_factor · Pre-Algebra – Integers, Decimals, and Fractions ·
gen=g6_greatest_common_factor

Name: _____ Date: _____

Find the GCF.

EASY

1. Find the GCF of 24, 28

MEDIUM

2. Find the GCF of 12, 48

HARD

3. Find the GCF of 60, 120, 20

Answer Key

1. **Easy:** 4

2. **Medium:** 12

3. **Hard:** 20

Least common multiple

pa_least_common_multiple · Pre-Algebra – Integers, Decimals, and Fractions ·
gen=g6_least_common_multiple

Name: _____ Date: _____

Find the LCM.

EASY

1. Find the LCM of 12, 6

MEDIUM

2. Find the LCM of 12, 47

HARD

3. Find the LCM of 17, 73, 113

Answer Key

1. **Easy:** 12

2. **Medium:** 564

3. **Hard:** 140233

Multiplying

pa_integers_multiplying · Pre-Algebra – Integers, Decimals, and Fractions · gen=g6_integer_multiply

Name: _____ Date: _____

Multiply.

EASY

1. $-5 \cdot (-10)$

MEDIUM

2. $13 \cdot 14$

HARD

3. $29 \cdot (-34)$

Answer Key

1. **Easy:** 50

2. **Medium:** 182

3. **Hard:** -986

Naming decimal places and rounding

pa_naming_decimal_places_and_rounding · Pre-Algebra – Integers, Decimals, and Fractions ·
gen=place_value_and_rounding

Name: _____ Date: _____

Name the place value or round.

EASY

1. What digit is in the hundredths place of 294.614?

MEDIUM

2. Round 93.712 to the nearest hundredth.

HARD

3. What digit is in the hundredths place of 935.407?

Answer Key

1. Easy: 1

2. Medium: 93.71

3. Hard: 0

Simplifying fractions

pa_simplifying_fractions · Pre-Algebra – Integers, Decimals, and Fractions ·
gen=simplifying_numeric_fractions

Name: _____ Date: _____

Simplify.

EASY

1. Simplify $\frac{12}{36}$.

MEDIUM

2. Simplify $\frac{80}{10}$.

HARD

3. Simplify $\frac{70}{10}$.

Answer Key

1. **Easy:** $\frac{1}{3}$

2. **Medium:** 8

3. **Hard:** 7

Writing numbers with words

pa_writing_numbers_with_words · Pre-Algebra – Integers, Decimals, and Fractions ·
gen=writing_numbers_with_words

Name: _____ Date: _____

Write the number in words.

EASY

1. Write 2033 in words.

MEDIUM

2. Write 2479 in words.

HARD

3. Write 2217 in words.

Answer Key

1. **Easy:** two thousand thirty-three
2. **Medium:** two thousand four hundred seventy-nine
3. **Hard:** two thousand two hundred seventeen

Graphing systems of equations

pa_graphing_systems_of_equations · Pre-Algebra – Linear Equations and Inequalities · gen=graph_system

Name: _____ Date: _____

Solve the system.

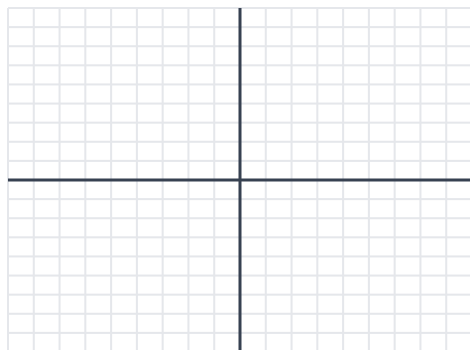
EASY

$$1. \begin{cases} y = -2x - 1 \\ y = -3x - 4 \end{cases}$$



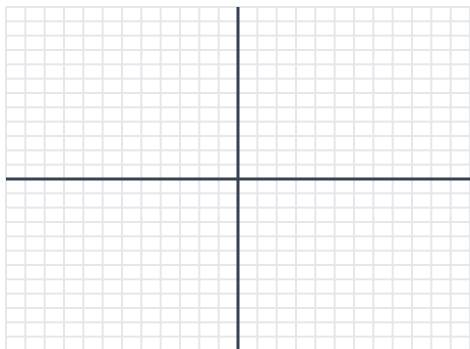
MEDIUM

$$2. \begin{cases} y = 2x - 9 \\ y = x - 8 \end{cases}$$



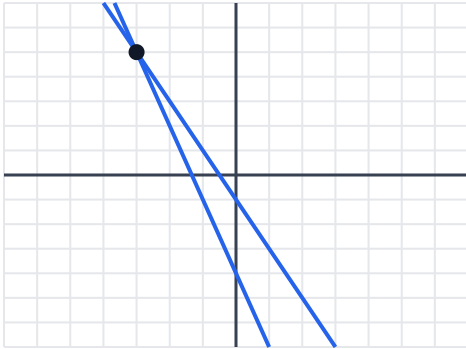
HARD

3.
$$\begin{cases} y = -4x + 29 \\ y = -9x + 59 \end{cases}$$

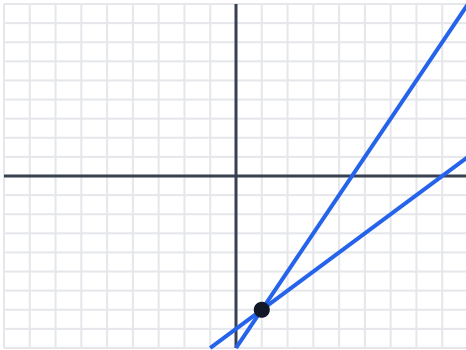


Answer Key

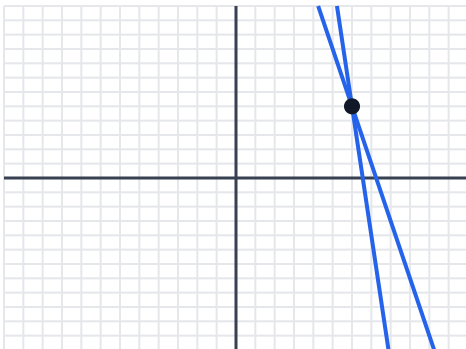
1. **Easy:** $(x, y) = (-3, 5)$



2. **Medium:** $(x, y) = (1, -7)$



3. **Hard:** $(x, y) = (6, 5)$



Plotting points

pa_plotting_points · Pre-Algebra – Linear Equations and Inequalities · gen=plotting_points

Name: _____ Date: _____

Plot the following points on the coordinate plane.

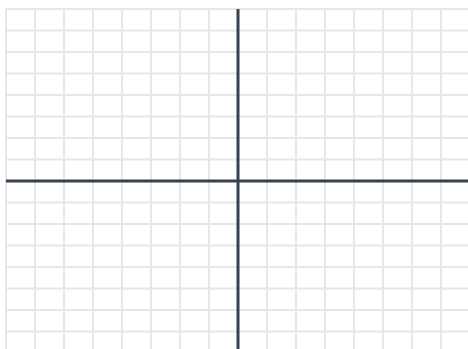
EASY

1. $(5, -1)$



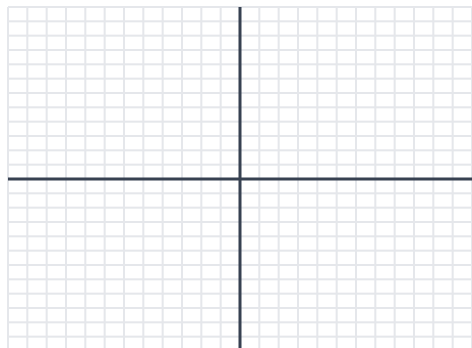
MEDIUM

2. $(4, -1)$



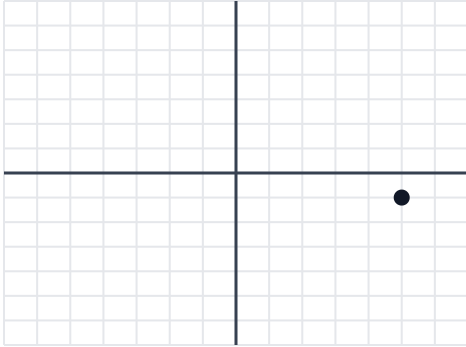
HARD

3. $(8, 2)$

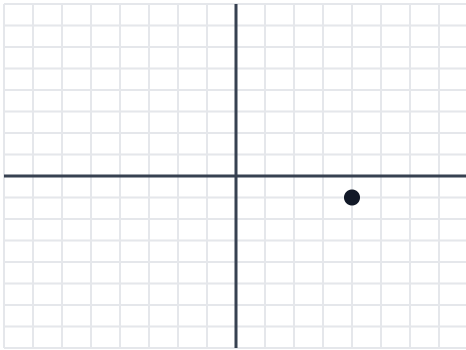


Answer Key

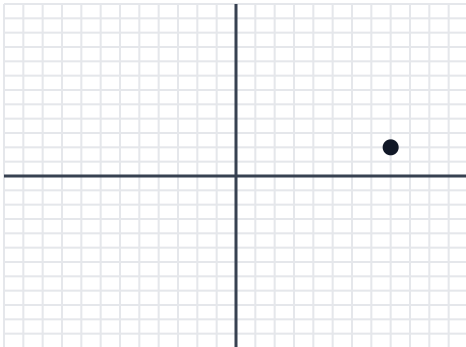
1. **Easy:** $(5, -1)$



2. **Medium:** $(4, -1)$



3. **Hard:** $(8, 2)$



Slope

pa_slope · Pre-Algebra – Linear Equations and Inequalities · gen=slope

Name: _____ Date: _____

Find the slope.

EASY

1. Find the slope of the line through $(5, -3)$ and $(7, -7)$.

MEDIUM

2. Find the slope of the line $10x + 2y = 6$.

HARD

3. Find the slope of the line $y - 1 = -5(x + 10)$.

Answer Key

1. **Easy:** -2
2. **Medium:** -5
3. **Hard:** -5

Solving systems of equations by substitution

pa_systems_substitution · Pre-Algebra – Linear Equations and Inequalities · gen=systems_substitution

Name: _____ Date: _____

Solve the system.

EASY

$$1. \begin{cases} y = 2x + 1 \\ -x + 4y = 25 \end{cases}$$

MEDIUM

$$2. \begin{cases} y = 6x + 18 \\ -8x - 9y = -38 \end{cases}$$

HARD

$$3. \begin{cases} y = 5x - 7 \\ -5x - 7y = -31 \end{cases}$$

Answer Key

1. **Easy:** $(x, y) = (3, 7)$

2. **Medium:** $(x, y) = (-2, 6)$

3. **Hard:** $(x, y) = (2, 3)$

Systems of equations word problems

pa_systems_word_problems · Pre-Algebra – Linear Equations and Inequalities · gen=wp_systems

Name: _____ Date: _____

Solve the problem.

EASY

1. Adult tickets cost \$12 and child tickets cost \$8. 66 tickets were sold for a total of \$680. How many adult tickets were sold?

MEDIUM

2. Adult tickets cost \$10 and child tickets cost \$6. 66 tickets were sold for a total of \$552. How many adult tickets were sold?

HARD

3. Adult tickets cost \$10 and child tickets cost \$6. 94 tickets were sold for a total of \$776. How many adult tickets were sold?

Answer Key

1. **Easy:** 38
2. **Medium:** 39
3. **Hard:** 53

Writing linear equations

pa_writing_linear_equations · Pre-Algebra – Linear Equations and Inequalities ·
gen=writing_linear_equations

Name: _____ Date: _____

Write the equation.

EASY

1. Write $12x - 4y = -20$ in slope-intercept form.

MEDIUM

2. Write the equation of the line with slope -2 and y -intercept 1 in slope-intercept form.

HARD

3. Write the equation of the line through $(6, -44)$ and $(0, -2)$ in slope-intercept form.

Answer Key

1. Easy: $y = 3x + 5$
2. Medium: $y = -2x + 1$
3. Hard: $y = -7x - 2$

Fractions, decimals, and percents

pa_fractions_decimals_and_percents · Pre-Algebra – Percents · gen=fractions_decimals_and_percents

Name: _____ Date: _____

Write the equivalent value.

EASY

1. Write $\frac{1}{5}$ as a percent.

MEDIUM

2. Write 20% as a fraction in simplest form.

HARD

3. Write 0.15 as a percent.

Answer Key

1. **Easy:** 20%

2. **Medium:** $\frac{1}{5}$

3. **Hard:** 15%

Markup, discount, and tax

pa_markup_discount_and_tax · Pre-Algebra – Percents · gen=wp_percent

Name: _____ Date: _____

Solve the problem.

EASY

1. Liam leaves a 15% tip on a \$109 bill.
How much is the tip?

MEDIUM

2. A \$103 purchase has 8% sales tax added. What is the total cost?

HARD

3. Jamie buys an item priced at \$192.89. It is on sale for 12.5% off. What is the sale price?

Answer Key

1. **Easy:** \$16.35
2. **Medium:** \$111.24
3. **Hard:** \$168.78

Simple and compound interest

pa_simple_and_compound_interest · Pre-Algebra – Percents · gen=wp_simple_and_compound_interest

Name: _____ Date: _____

Solve the problem.

EASY

1. Uri invests \$200 at 6% simple interest for 5 years. How much interest is earned?

MEDIUM

2. Marcus invests \$1000 at 9% simple interest for 3 years. How much interest is earned?

HARD

3. Nina invests \$1000 at 7% interest compounded semiannually for 8 years. What is the balance at the end of the term?

Answer Key

1. **Easy:** \$60
2. **Medium:** \$270
3. **Hard:** \$1733.99

Angle relationships

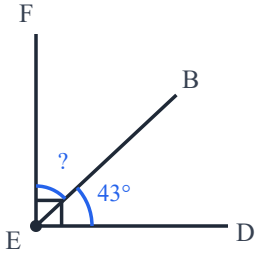
pa_angle_relationships · Pre-Algebra – Plane Figures · gen=geo_angle_relationships

Name: _____ Date: _____

Find the angle measure.

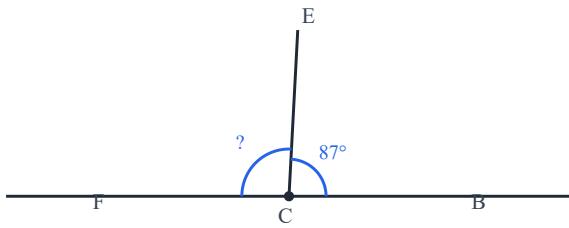
EASY

1. The diagram shows complementary angles. Find the measure of the unmarked angle.



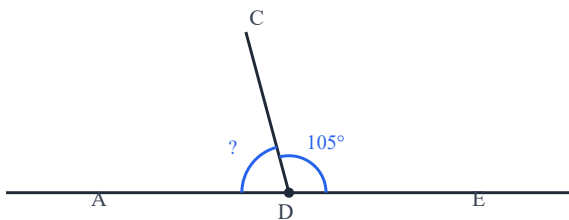
MEDIUM

2. The diagram shows supplementary angles on a straight line. Find the measure of the unmarked angle.



HARD

3. The diagram shows supplementary angles on a straight line. Find the measure of the unmarked angle.



Answer Key

1. Easy: 47°
2. Medium: 93°
3. Hard: 75°

Area of triangles and quadrilaterals

pa_area_of_triangles_and_quadrilaterals · Pre-Algebra – Plane Figures ·
gen=geo_triangles_and_quadrilaterals_area

Name: _____ Date: _____

Find the area.

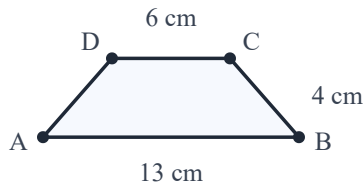
EASY

1. Find the area of the rectangle.



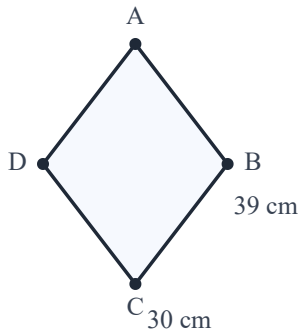
MEDIUM

2. Find the area of the trapezoid.



HARD

3. Find the area of the kite.



Answer Key

1. Easy: 36 cm^2

2. Medium: 38 cm^2

3. Hard: 585 cm^2

Circles

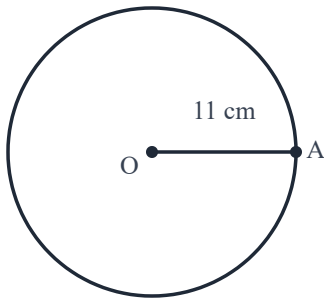
pa_circles · Pre-Algebra – Plane Figures · gen=geo_circle_measure

Name: _____ Date: _____

Find the measure.

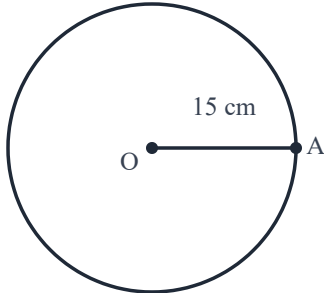
EASY

1. Find the circumference of a circle with radius 11 cm.



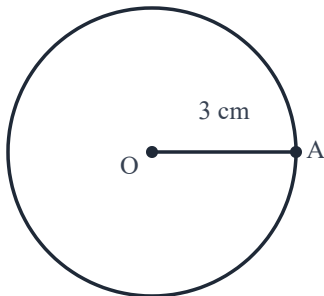
MEDIUM

2. Find the area of a circle with radius 15 cm.



HARD

3. Find the area of a circle with radius 3 cm.



Answer Key

1. **Easy:** 22π cm
2. **Medium:** 225π cm²
3. **Hard:** 9π cm²

Drawing and measuring angles

pa_drawing_and_measuring_angles · Pre-Algebra – Plane Figures · gen=geo_angles

Name: _____ Date: _____

Measure or draw the angle.

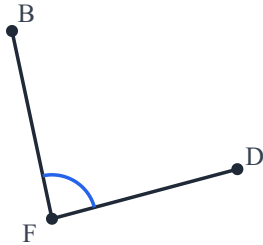
EASY

1. Ray FA is drawn. Draw ray FC so that $m\angle AFC = 40^\circ$.



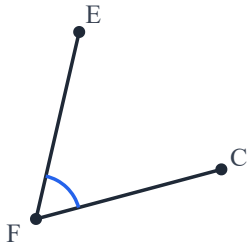
MEDIUM

2. Find $m\angle DFB$ using the diagram.



HARD

3. Find $m\angle CFE$ using the diagram.



Answer Key

1. Easy: 40°
2. Medium: 87°
3. Hard: 62°

Quadrilaterals

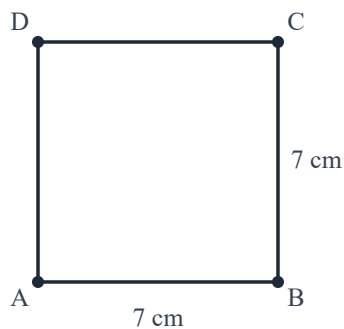
pa_quadrilaterals · Pre-Algebra – Plane Figures · gen=geo_quadrilateral_area

Name: _____ Date: _____

Find the measure.

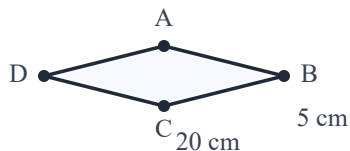
EASY

1. Find the area of the square.



MEDIUM

2. Find the area of the kite.



HARD

3. Find the area of the rectangle.



Answer Key

1. **Easy:** 49 cm^2
2. **Medium:** 50 cm^2
3. **Hard:** 300 cm^2

Transformations

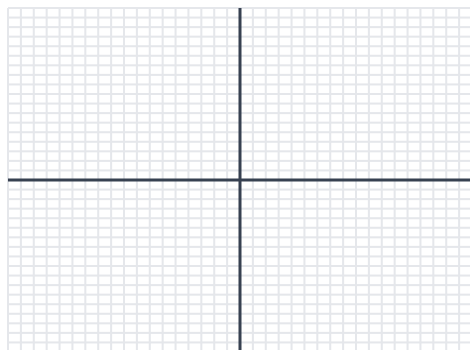
pa_transformations · Pre-Algebra – Plane Figures · gen=graph_transformations

Name: _____ Date: _____

Graph the transformation.

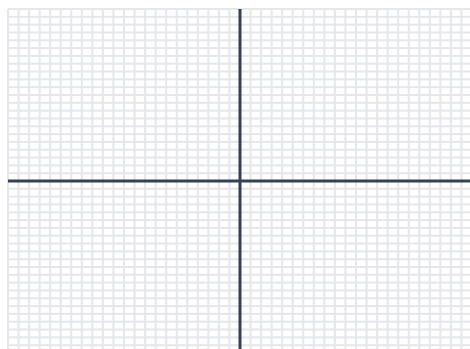
EASY

1. Graph the transformation of $f(x) = x^2$ to $y = (x - 0)^2 + 0$.



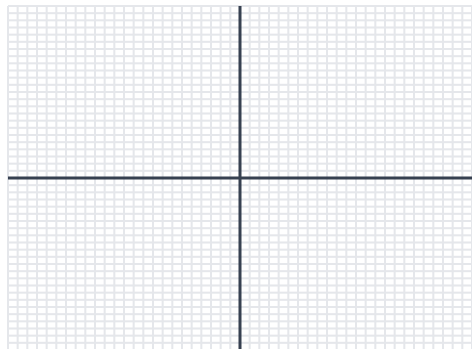
MEDIUM

2. Graph the transformation of $f(x) = x^2$ to $y = (x - 6)^2 + 4$.



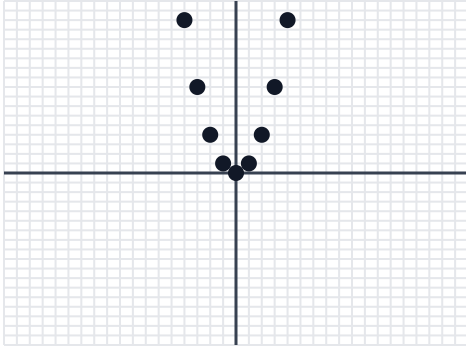
HARD

3. Graph the transformation of $f(x) = x^2$ to $y = (x + 7)^2 + 6$.

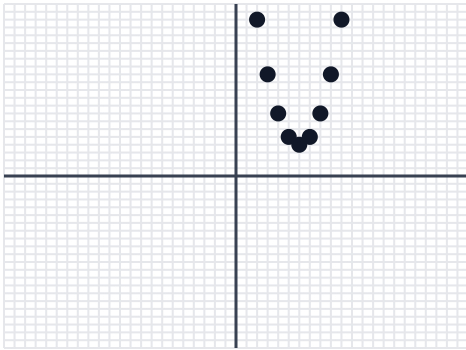


Answer Key

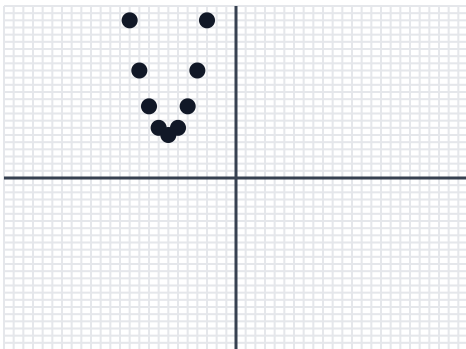
1. **Easy:** $y = (x - 0)^2 + 0$



2. **Medium:** $y = (x - 6)^2 + 4$



3. **Hard:** $y = (x + 7)^2 + 6$



Triangles

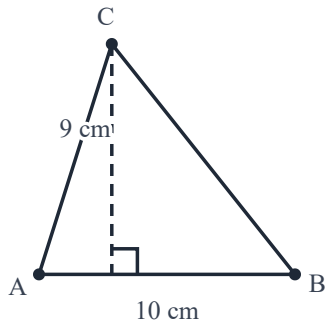
pa_plane_figures_triangles · Pre-Algebra – Plane Figures · gen=geo_triangle_area

Name: _____ Date: _____

Find the measure.

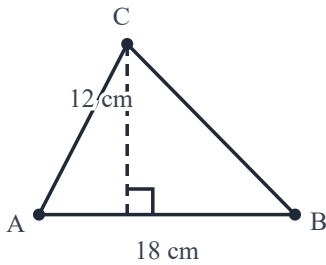
EASY

1. Find the area of $\triangle ABC$.



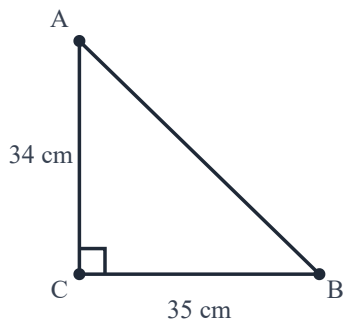
MEDIUM

2. Find the area of $\triangle ABC$.



HARD

3. Find the area of $\triangle ABC$.



Answer Key

1. **Easy:** 45 cm^2
2. **Medium:** 108 cm^2
3. **Hard:** 595 cm^2

Checking for a proportion

pa_checking_for_a_proportion · Pre-Algebra – Proportions and Similarity · gen=solving_proportions

Name: _____ Date: _____

Determine if the ratios form a proportion.

EASY

1. $\frac{5}{5} = \frac{x}{8}$

MEDIUM

2. $\frac{2}{2} = \frac{x}{10}$

HARD

3. $\frac{8}{4} = \frac{x}{7}$

Answer Key

1. **Easy:** 8

2. **Medium:** 10

3. **Hard:** 14

Proportions word problems

pa_proportions_word_problems · Pre-Algebra – Proportions and Similarity · gen=wp_proportion

Name: _____ Date: _____

Solve the problem.

EASY

1. If 4 apples cost \$1.25, how much do 10 apples cost at the same rate?

MEDIUM

2. If 3 notebooks cost \$1.25, how much do 8 notebooks cost at the same rate?

HARD

3. If 5 apples cost \$1.00, how much do 9 apples cost at the same rate?

Answer Key

1. Easy: \$3

2. Medium: \$3

3. Hard: \$2

Similar figures

pa_similar_figures · Pre-Algebra – Proportions and Similarity · gen=wp_similar_figures

Name: _____ Date: _____

Find the missing measure.

EASY

1. Two similar triangles have a scale factor of 2:1. The smaller triangle has a side of 5 cm. What is the corresponding side in the larger triangle?

MEDIUM

2. Two similar triangles have a scale factor of 4:1. The smaller triangle has a side of 9 cm. What is the corresponding side in the larger triangle?

HARD

3. Two similar triangles have a scale factor of 4:1. The smaller triangle has a side of 6 cm. What is the corresponding side in the larger triangle?

Answer Key

1. **Easy:** 10
2. **Medium:** 36
3. **Hard:** 24

Similar figures word problems

pa_similar_figures_word_problems · Pre-Algebra – Proportions and Similarity · gen=wp_similar_figures

Name: _____ Date: _____

Solve the problem.

EASY

1. Two similar triangles have a scale factor of 3:1. The smaller triangle has a side of 4 cm. What is the corresponding side in the larger triangle?

MEDIUM

2. Two similar triangles have a scale factor of 4:1. The smaller triangle has a side of 3 cm. What is the corresponding side in the larger triangle?

HARD

3. Two similar triangles have a scale factor of 3:1. The smaller triangle has a side of 5 cm. What is the corresponding side in the larger triangle?

Answer Key

1. **Easy:** 12
2. **Medium:** 12
3. **Hard:** 15

The Pythagorean Theorem

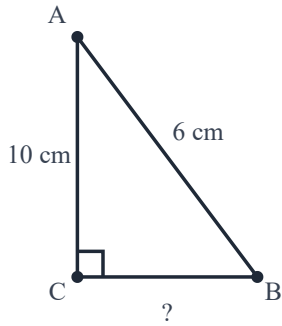
pythagorean_theorem · Pre-Algebra – Right Triangles · gen=geo_pythagorean_theorem

Name: _____ Date: _____

Find the missing side.

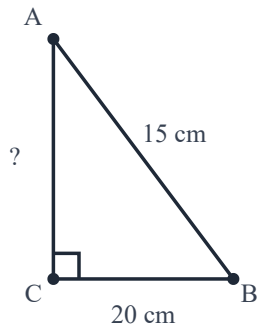
EASY

1. In right $\triangle ABC$, find BC .



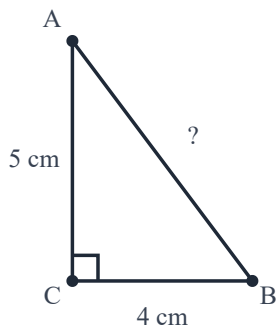
MEDIUM

2. In right $\triangle ABC$, find AC .



HARD

3. In right $\triangle ABC$, find AB .



Answer Key

1. **Easy:** 8 cm
2. **Medium:** 25 cm
3. **Hard:** 3 cm

Classifying, volume, and surface area

pa_classifying_volume_and_surface_area · Pre-Algebra – Solid Figures · gen=geo_solid_volume_surface

Name: _____ Date: _____

Find the volume or surface area.

EASY

1. Find the surface area of a $5 \times 3 \times 9$ cm rectangular prism.

MEDIUM

2. Find the surface area of a $17 \times 16 \times 7$ cm rectangular prism.

HARD

3. Find the volume of a $27 \times 29 \times 6$ cm rectangular prism.

Answer Key

1. **Easy:** 174 cm^2
2. **Medium:** 1006 cm^2
3. **Hard:** 4698 cm^3